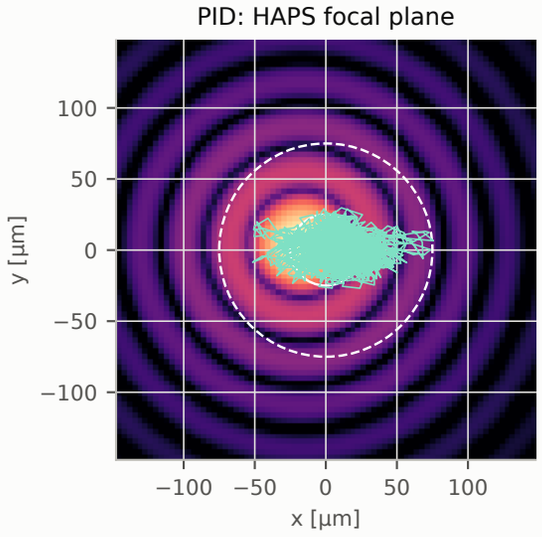
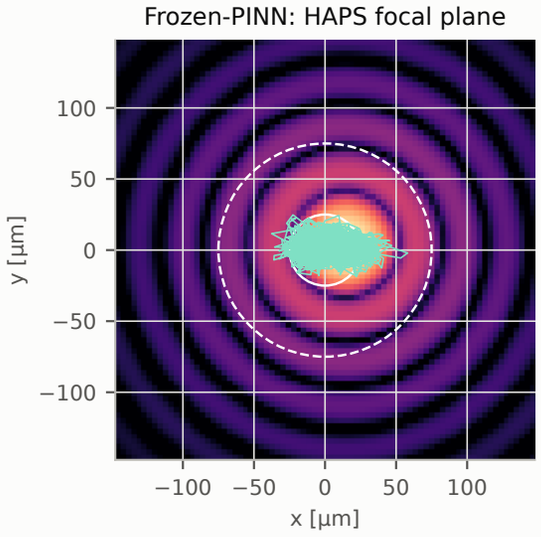
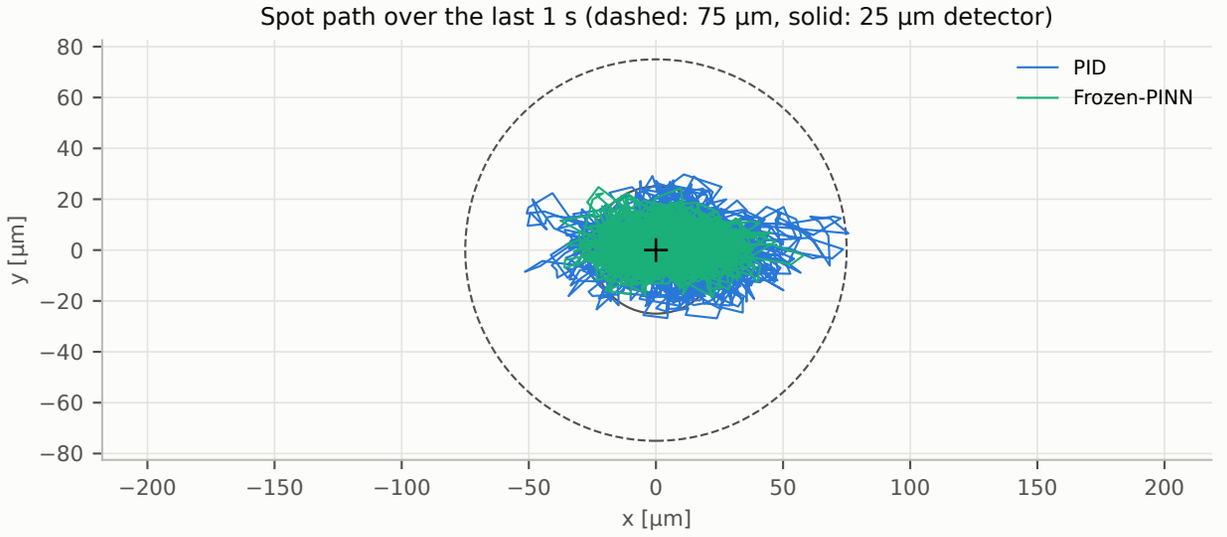
PAT + 64-QAM link | Fast lateral motion, 6-14 m/s | t = 10.00 s

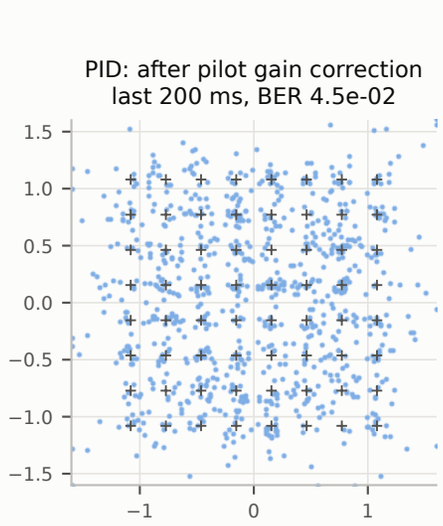
Spot: HAPS-receiver focal plane (log intensity) shifted by the residual. Symbols: 64-QAM at the operating SNR, pilot block every 0.5 ms.



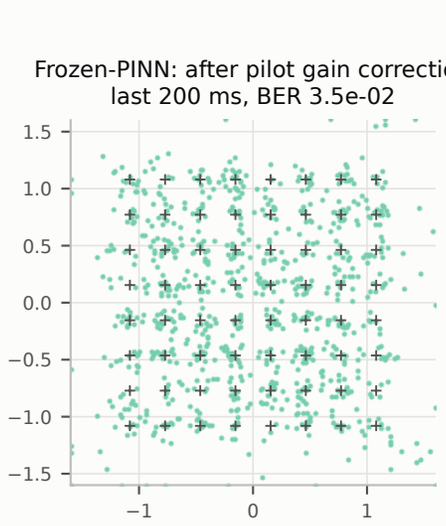




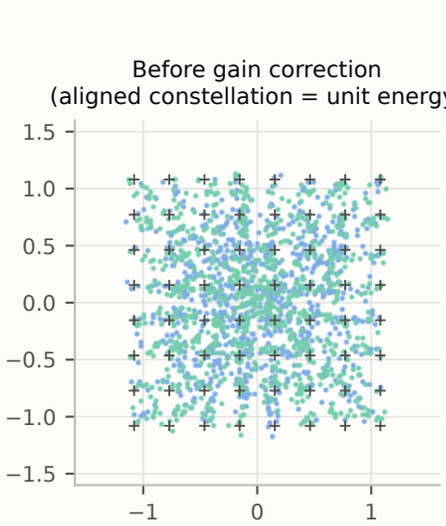
PID: after pilot gain correction
last 200 ms, BER 4.5e-02



Frozen-PINN: after pilot gain correction
last 200 ms, BER 3.5e-02



Before gain correction
(aligned constellation = unit energy)



t = 10.00 s
Fast lateral motion, 6-14 m/s
25 µm detector, aligned SNR
24.1 dB (FEC + 3 dB)

PID (last 200 ms)
- RMS offset 17.7 µm
- mean power 0.714
- outage 42.6 %
- BER 4.5e-02

Frozen-PINN (last 200 ms)
- RMS offset 15.4 µm
- mean power 0.786
- outage 23.7 %
- BER 3.5e-02

